

Explicit Conformal Restriction Bounds for the Exact- U_c Polar Fredholm Determinant

Anonymous

August 2026

Abstract

For the unchanged exact- U_c polar Fredholm determinant, we make the normalized conformal restriction constants in its nuclear Taylor factorization completely explicit. The frozen operator domains are thin stadiums of radius 10^{-3} , while the proof-only inner stadiums have radius 6×10^{-4} . Domain monotonicity of the curvature- -1 Poincare metric bounds a midpoint-projection-point path by $D_* = 500\pi + \log 4$. Hyperbolic invariance of the normalized Riemann maps then gives

$$r_L = r_R \leq \tanh(D_*/2) =: r_* < 1.$$

A 4096-bit outward Arb certificate resolves $1 - r_* = 3.2418512480136249798 \dots \times 10^{-683}$, a gap that ordinary floating-point arithmetic rounds away. Inserting the bound into the inherited two-stream Fredholm coefficient estimate yields the fully numerical same-determinant envelope

$$|D_{\text{pol}}(s)| \leq \exp(3.45 \times 10^{689} + 4.20 \times 10^{682}(1 + |s|)^2).$$

The result closes an explicit-proof-constant obligation. It does not compute the exact conformal ratios, determine the true growth type, evaluate a determinant root, or provide arithmetic divisor or quantization evidence.

1 Frozen determinant and main result

Let $u = U_c$ be the real root of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad \rho = u - 1.$$

The inherited real dynamics is the exact two-full-branch polar map on $I_L = [-\pi/2, 0]$ and $I_R = [0, \pi/2]$. Its complex transfer family acts on

$$X = A(U_L) \oplus A(U_R), \quad B = \{(v_L, v_R) : v_L(0) = v_R(0)\},$$

where U_L, U_R are the radius- 10^{-3} stadiums about the two branch intervals. With the inherited inverse branches and common logarithmic weight, the operator is

$$(\mathcal{L}_s v)_j(z) = e^{s\ell(z)} [v_L(\phi_L(z)) + v_R(\phi_R(z))]. \quad (1)$$

The preceding stage proves that $\mathcal{L}_s|_B$ is nuclear of order zero and that

$$D_{\text{pol}}(s) = \det_{\text{Fr}}(I - \mathcal{L}_s|_B) \quad (2)$$

is entire. Equation (2), its roof clock, and its signed trace convention are fixed throughout this paper.

For $R = 10^{-3}$ and $r_0 = 6 \times 10^{-4}$, define

$$U_\sigma = \{z : \text{dist}(z, I_\sigma) < R\}, \quad V_\sigma = \{z : \text{dist}(z, I_\sigma) < r_0\}. \quad (3)$$

Normalize $h_\sigma : \mathbb{D} \rightarrow U_\sigma$ by

$$h_L(0) = -\frac{\pi}{4}, \quad h_R(0) = \frac{\pi}{4}, \quad h'_\sigma(0) > 0,$$

and put

$$r_\sigma = \max_{z \in \overline{V}_\sigma} |h_\sigma^{-1}(z)|. \quad (4)$$

Compact containment makes the maximum in (4) well defined.

Theorem 1 (Explicit conformal and determinant bounds). *Set*

$$D_* = 500\pi + \log 4, \quad r_* = \tanh(D_*/2). \quad (5)$$

Then

$$r_L = r_R \leq r_* < 1. \quad (6)$$

Moreover, the same determinant (2) satisfies

$$|D_{\text{pol}}(s)| \leq \exp\left(3.45 \times 10^{689} + 4.20 \times 10^{682}(1 + |s|)^2\right) \quad (s \in \mathbb{C}). \quad (7)$$

The theorem gives an explicit upper envelope only. It does not identify the exact conformal modulus or the true entire-function type.

2 A hyperbolic path inside each stadium

Use the curvature -1 Poincare convention

$$\lambda_{\mathbb{D}}(w) = \frac{2}{1 - |w|^2}, \quad d_{\mathbb{D}}(0, w) = 2 \operatorname{artanh} |w|. \quad (8)$$

Fix σ and $z \in \overline{V}_\sigma$. Let p be the Euclidean projection of z onto I_σ . The path used in the proof first travels along the real interval from its midpoint c_σ to p , and then follows the line segment from p to z .

Every disk $B(x, R)$, with $x \in I_\sigma$, lies in U_σ . Hyperbolic density monotonicity therefore gives

$$\lambda_{U_\sigma}(x) \leq \frac{2}{R}.$$

The branch interval has length $L = \pi/2$, and its midpoint is at distance at most $L/2$ from p . Consequently

$$d_{U_\sigma}(c_\sigma, p) \leq \frac{2}{R} \frac{L}{2} = 500\pi. \quad (9)$$

The second segment lies in $B(p, R)$, including when $z \in \partial V_\sigma$, because $|z - p| \leq r_0 < R$. Disk distance and domain monotonicity give

$$d_{U_\sigma}(p, z) \leq 2 \operatorname{artanh} \frac{r_0}{R} = \log \frac{R + r_0}{R - r_0} = \log 4. \quad (10)$$

Thus

$$d_{U_\sigma}(c_\sigma, z) \leq D_*. \quad (11)$$

Since h_σ is a hyperbolic isometry, (8) and (11) imply

$$|h_\sigma^{-1}(z)| \leq \tanh(D_*/2) = r_*.$$

Translation by $\pi/2$ maps the left normalized stadium pair to the right one, so uniqueness of the normalized Riemann map also gives $r_L = r_R$.

The small gap below one must be evaluated stably. With $t = e^{-D_*}$, write

$$r_* = \frac{1-t}{1+t}, \quad \delta_* = 1 - r_* = \frac{2t}{1+t}, \quad \beta_* = -\log r_* = \log \frac{1+t}{1-t}. \quad (12)$$

These formulas preserve positivity without subtracting two ordinary-precision numbers that both display as one.

3 From the conformal gap to numerical growth constants

The inherited matching-space factorization has two geometric rank-one streams. Its weight satisfies

$$W(s) \leq e^{L_\ell |s|}, \quad L_\ell = \frac{103}{125}. \quad (13)$$

For one stream, the exact elementary-symmetric identity contains $\prod_{h=1}^k (1 - r_\sigma^h)^{-1}$. Since $r_\sigma \leq r_*$, each inverse factor is at most δ_*^{-1} . Splitting rank q as $q = k + (q - k)$ between the two streams and retaining the inherited Hadamard factor gives the determinant coefficients

$$|a_q(s)| \leq q^{q/2}(q+1) \left(\frac{e^{L_\ell |s|}}{\delta_*} \right)^q r_*^{q^2/4 - q/2}. \quad (14)$$

This is a bound for the same canonical determinant, not for a finite matrix or a separately glued product.

Choose $\theta = 1/4096$ and define

$$b_\theta = -\log \delta_* + \frac{\beta_*}{2} + \frac{1}{2} \left(\log \frac{1}{\theta \beta_*} - 1 \right) + \log 2. \quad (15)$$

The elementary inequalities

$$\log q \leq \theta \beta_* q + \log \frac{1}{\theta \beta_*} - 1, \quad \log(q+1) \leq q \log 2 \quad (16)$$

turn (14) into

$$|a_q(s)| \leq \exp \left[-\frac{\beta_*(1-2\theta)}{4} q^2 + (L_\ell |s| + b_\theta) q \right]. \quad (17)$$

Completing the square and summing the shifted Gaussian over the integer lattice yields

$$|D_{\text{pol}}(s)| \leq \left(1 + \sqrt{\frac{4\pi}{\beta_*(1-2\theta)}} \right) \exp \left(\frac{(L_\ell |s| + b_\theta)^2}{\beta_*(1-2\theta)} \right). \quad (18)$$

Finally, write $L_\ell |s| + b_\theta = L_\ell(1 + |s|) + (b_\theta - L_\ell)$ and apply $(x+y)^2 \leq 2x^2 + 2y^2$. The resulting formulas are

$$C_0 = \log \left(1 + \sqrt{\frac{4\pi}{\beta_*(1-2\theta)}} \right) + \frac{2(b_\theta - L_\ell)^2}{\beta_*(1-2\theta)}, \quad (19)$$

$$C_1 = \frac{2L_\ell^2}{\beta_*(1-2\theta)}. \quad (20)$$

4 Outward interval certificate

The companion program evaluates (5), (12), and (15)–(20) with 4096-bit Arb balls. Table 1 lists the central displayed values; the machine-readable artifact retains the outward radii.

Quantity	Certified displayed value
D_*	1572.18262115601650985015615588...
δ_*	$3.24185124801362497983758530 \dots \times 10^{-683}$
β_*	$3.24185124801362497983758530 \dots \times 10^{-683}$
b_θ	2361.58624122710446397702901049...
C_0	$3.43996102886264722070369148 \dots \times 10^{689}$
C_1	$4.19086282028397353977286883 \dots \times 10^{682}$

Table 1: Target-free outward interval calculation. The decimal ceilings 3.45×10^{689} and 4.20×10^{682} lie strictly above the certified balls for C_0 and C_1 .

The artifact also checks the exact rational geometry

$$\frac{r_0}{R} = \frac{3}{5}, \quad \frac{R + r_0}{R - r_0} = 4, \quad \frac{2}{R} \pi = 500\pi,$$

freezes ‘python-flint 0.9.0’ and FLINT ‘3.6.0’, records source hashes, and reproduces byte for byte. Its data firewall marks every prime table, Riemann-zero table, zeta or xi evaluation, Fredholm evaluation, conformal solver, and fitting flag as false.

The certificate does not validate a numerical approximation of a Riemann map. It validates only scalar consequences of the analytic path theorem.

5 Limitations and conclusion

The explicit conformal estimate closes a narrow but necessary reproducibility gap in the LOG-0001 growth theorem. Every constant in the quadratic exponential envelope can now be evaluated from the frozen stadium geometry and the inherited logarithmic-weight bound. No fitted conformal map or external arithmetic data enters the calculation.

The estimate is intentionally elementary. The outer stadium is long relative to its radius, so a path from its midpoint to an endpoint has hyperbolic length of order 10^3 . The resulting gap below one is therefore of order 10^{-683} , and the determinant constants are correspondingly large. These numbers certify finiteness; they do not describe the true conformal radius or Fredholm growth type.

The Route-A status is unchanged. The construction has a genuine entire dynamical determinant and partial global analytic control, but it still has no log-prime primitive-orbit law, von-Mangoldt trace weights, sharp divisor asymptotic, completed-xi identity, or natural quantization. In particular, the present theorem gives no determinant root and no Route-B authorization.

A smallest follow-up is a structural precheck for lower growth: one must name an explicit coefficient or signed trace contribution together with a cancellation-safe mechanism that transfers it to a lower bound for the same determinant. Without such a mechanism, the lower-growth task is not testable and should not be replaced by numerical root fitting.

A Auxiliary proofs

Lemma 2 (Logarithmic linearization). *For $a > 0$ and $x > 0$,*

$$\log x \leq ax + \log(1/a) - 1.$$

Proof. The function $\log x - ax$ reaches its global maximum at $x = 1/a$, where its value is $\log(1/a) - 1$. $\square$

Lemma 3 (Shifted Gaussian lattice). *For $\alpha > 0$ and $c \in \mathbb{R}$,*

$$\sum_{q \in \mathbb{Z}} e^{-\alpha(q-c)^2} \leq 1 + \sqrt{\frac{\pi}{\alpha}}.$$

Proof. Poisson summation shows that the shifted sum is maximized when c is an integer, because every Fourier coefficient of the Gaussian is positive. At an integral shift,

$$\sum_{q \in \mathbb{Z}} e^{-\alpha q^2} = 1 + 2 \sum_{q \geq 1} e^{-\alpha q^2}.$$

Since $x \mapsto e^{-\alpha x^2}$ decreases on $[0, \infty)$, each term in the positive tail is bounded by the integral over the preceding unit interval. Thus

$$2 \sum_{q \geq 1} e^{-\alpha q^2} \leq 2 \int_0^\infty e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}.$$

$\square$

Proposition 4 (Coefficient summation). *Suppose $a_0 = 1$ and for $q \geq 1$,*

$$|a_q| \leq \exp(-\alpha q^2 + Bq), \quad \alpha > 0.$$

Then

$$\left| \sum_{q \geq 0} a_q \right| \leq \left(1 + \sqrt{\frac{\pi}{\alpha}} \right) \exp\left(\frac{B^2}{4\alpha} \right).$$

Proof. Extend the positive majorant from $q \geq 0$ to all integer q , complete the square, and apply the preceding lemma with $c = B/(2\alpha)$. $\square$

The passage from finite Taylor truncations to D_{pol} is inherited from the order-zero nuclear theorem. The present paper changes only the numerical majorant and does not introduce a second determinant limit.

References

No external references are cited in this self-contained stage report. The argument uses the frozen determinant established in the preceding project stage and standard facts about the Poincare metric, Riemann maps, Fredholm coefficients, and Gaussian sums.